

REPUBLIC OF SOUTH AFRICA

PATENTS ACT, 1978

APPLICATION FOR A PATENT AND ACKNOWLEDGEMENT OF RECEIPT

[Section 30 (1)-Regulation 22]

The granting of a patent is hereby requested by the undermentioned applicant on the basis of the present application.

Official Application No.			Applicant's or Agent's Reference	
21	01	2025/10600	6017LPS	

71	Full Name(s) of Applicant(s)			
VISHWAKARMA INSTITUTE OF TECHNOLOGY 666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India				

54	Title of invention			
AN IRRIGATION SYSTEM FOR AUTOMATED PLANT WATERING BASED ON SOIL MOISTURE CONDITIONS				

The applicant claims priority as set out on the accompanying Form P.2. The earliest priority claimed is				DATE:
COUNTRY:		NUMBER:		

This application is for a patent of addition to patent application No.				
21	01			

This application is a fresh application in terms of section 37 and based on Application No.				
21	01			

This application is accompanied by:				
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X	1.	A single copy of a complete specification of 10 pages.
X	2.	Drawings of 1 sheet(s).
X	3.	Publication particulars and abstract(Form P8)
X	4.	A copy of a figure of the drawing (if any) for the abstract
X	5.	Assignment of invention
	6.	Certified priority document(s)
	7.	Translation(s) of the priority document(s)
	8.	Assignment of priority rights
	9.	A copy of the Form P.2 and the specification of S.A Patent Application (if applicable).
X	10.	A declaration and power of attorney on Form P3
X	11.	Statement on the use of indigenous Biological Resource, Genetic Resource, Traditional Knowledge or Use on Form P26

74	Address of Service:			
Wolmarans and Susan Inc. 337 Surrey Avenue, Randburg, Gauteng, 2194 SOUTH AFRICA				

Dated this 9th day of December 2025

Digitally signed by : Leon Pierre Susan

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Signature of Applicant(s)

This is returned to the applicant's
address for service as proof of lodging.

RECEIVED
Official Date Stamp
..... Registrar of Patents

REPUBLIC OF SOUTH AFRICA

REGISTER OF PATENTS

PATENTS ACT, 1978

Official application No.		Lodging date: Provisional		Acceptance date	
21	01	2025/10600		22	
International classification		Lodging date: Complete		Granted date	
51	A01G		23	2025/12/09	
71 Full name(s) of applicant(s)/Patentee(s):					
VISHWAKARMA INSTITUTE OF TECHNOLOGY 666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India					
71 Applicant substituted:				Date registered	
71 Assignee(s):				Date registered	
72 Full name(s) of inventor(s):					
JAHANGIR, Umar Tabrez KHONDRE, Shruti JALGAONKAR, Sahil Sanjay KALME, Rushab CHAUDHARI, Prasad B. CHAVHAN, Pranali G. SURYAWANSHI, Amol					
Priority claimed:		Country		Number	
54 Title of invention					
AN IRRIGATION SYSTEM FOR AUTOMATED PLANT WATERING BASED ON SOIL MOISTURE CONDITIONS					
Address of applicant(s)/patentee(s):					
666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037 INDIA					
74 Address for service					
Wolmarans and Susan Inc. 337 Surrey Avenue, Randburg, Gauteng, 2194 SOUTH AFRICA Reference No. 6017LPS					
61 Patent of addition No.				Date of any change	
Fresh application based on.				Date of any change	

REPUBLIC OF SOUTH AFRICA
PATENTS ACT, 1978
COMPLETE SPECIFICATION
[Section 30(1) – Regulation 28]

OFFICIAL APPLICATION NO.

21	01	2025/10600
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LODGING DATE

22	2025/12/09
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INTERNATIONAL CLASSIFICATION

51	A01G
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FULL NAME(S) OF APPLICANT(S)

71	VISHWAKARMA INSTITUTE OF TECHNOLOGY 666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India
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FULL NAME(S) OF INVENTORS(S)

72	1. JAHANGIR, Umar Tabrez 2. KHONDRE, Shruti 3. JALGAONKAR, Sahil Sanjay 4. KALME, Rushab 5. CHAUDHARI, Prasad B. 6. CHAVHAN, Pranali G. 7. SURYAWANSHI, Amol
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TITLE OF INVENTION

54	AN IRRIGATION SYSTEM FOR AUTOMATED PLANT WATERING BASED ON SOIL MOISTURE CONDITIONS
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REPUBLIC OF SOUTH AFRICA
PATENTS ACT, 1978
PUBLICATION PARTICULARS AND ABSTRACT
[Section 32(3)(a) – Regulation 2291)(g) AND 31]

OFFICIAL APPLICATION NO.		LODGING DATE	ACCEPTANCE DATE
21	01	22	2025/12/09
2025/10600			

INTERNATIONAL CLASSIFICATION	NOT FOR PUBLICATION
51 A01G	CLASSIFIED BY: Wolmarans and Susan Inc.

FULL NAME(S) OF APPLICANT(S)	
71	VISHWAKARMA INSTITUTE OF TECHNOLOGY 666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India

FULL NAME(S) OF INVENTORS(S)	
72	1. JAHANGIR, Umar Tabrez 2. KHONDRE, Shruti 3. JALGAONKAR, Sahil Sanjay 4. KALME, Rushab 5. CHAUDHARI, Prasad B. 6. CHAVHAN, Pranali G. 7. SURYAWANSHI, Amol

EARLIEST PRIORITY CLAIMED		
COUNTRY	NUMBER	DATE
33	31	32

TITLE OF INVENTION	
54	AN IRRIGATION SYSTEM FOR AUTOMATED PLANT WATERING BASED ON SOIL MOISTURE CONDITIONS

57	The present invention is related to an irrigation system for automated plant watering based on soil moisture conditions. The present invention discloses an irrigation system that provides automated, real-time water management for agricultural applications. The system comprises a soil moisture sensor, microcontroller, relay-operated water pump, and a Real-Time Clock (RTC) module for timestamped event logging. When soil dryness is detected, the microcontroller activates the pump to deliver water at a controlled flow rate, and automatically deactivates it upon achieving adequate moisture levels. Each irrigation event is recorded with precise time, duration, and water usage, which is calculated based on pump flow rate and runtime. The data is transmitted to a Python-based analytics dashboard that visualizes real-time soil conditions, water usage patterns, and long-term irrigation history. Unlike conventional systems, the invention enables accurate water conservation, predictive irrigation insights, and remote scalability through IoT modules, thereby reducing manual effort, preventing over- or under-watering, and enhancing crop productivity through sustainable and intelligent water management.
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